

SUPPLEMENTARY MATERIAL

ClinicalTensorSepsis: a harmonized multi-cohort temporal resource for Sepsis-3 research

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Abstract

This supplementary document provides additional methodological details and quality-control analyses related to the development of ClinicalTensorSepsis, a harmonized multi-cohort temporal resource for Sepsis-3 research. It describes the procedures used for source-specific cohort selection, suspected-infection screening, Sepsis-3 adjudication, hourly physiological tensor construction, missing-data imputation and cross-database representation transport. Particular attention is given to timing uncertainty, measurement provenance, organ-dysfunction assessment and incomplete clinical follow-up. Supplementary figures summarize observation coverage, imputation performance, transport and preservation diagnostics, hospital-level variation, cohort composition and sensitivity analyses. Additional numerical results are provided in the accompanying supplementary tables.

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1 Data sources and study design

1.1 Source databases and processing sequence

ClinicalTensorSepsis was constructed using MIMIC-IV version 3.1, the CareVue subset of MIMIC-III version 1.4, and eICU-CRD version 2.0 [1–3]. Each database was processed separately for cohort selection, infection screening, phenotype annotation, temporal extraction, measurement cleaning, Sepsis-3 adjudication, tensor construction, imputation and quality control. Cross-database representation learning and transport were performed after the source-specific datasets had been prepared. This sequence ensured that cohort decisions were based on recorded source data rather than imputed values.

A shared feature registry specified the clinical concepts, units and tensor positions used across the databases. Separate mappings described how each concept was identified in each source. Variables assigned to the same channel therefore required compatible definitions and units, while differences in infection assessment, timing precision and treatment records were retained. When a database did not contain sufficient information for a channel, that channel remained missing. Variables were not required to be available in every database.

1.2 Patient identity and temporal conventions

The analysis included the first ICU stay for each patient in each source database. This stay was identified before the remaining eligibility criteria were applied, using the chronology rules described below. Source identifiers were retained, and each stay was assigned a database-qualified key to distinguish identical numeric identifiers from different databases. The same patient order was used across cohort metadata, tensors, outcomes, data partitions and representations. Patients were not linked across databases, and different identifier systems or database releases were not taken as evidence that the MIMIC cohorts were independent.

Events were organized by their recorded clinical time rather than by the time they were entered into the database. The resulting trajectories therefore represent physiological measurements assembled retrospectively. In eICU, relative minute offsets were converted to a common reference point for interval calculations; this reference point did not represent a calendar date. Temporal metadata were expressed as integer minutes from ICU admission, while tensor hours were defined relative to the adjudicated time of sepsis onset.

Hospital mortality was retained as an outcome annotation but was not used to train the imputation model, encoder or transport mapping, or to choose their settings. Source-specific exclusions, candidate-level decisions and record-level quality-control statuses were recorded separately to describe cohort retention and measurement availability.

2 Base cohort construction

2.1 MIMIC-IV and MIMIC-III CareVue

For each patient, ICU stays were ordered by admission time, with the stay identifier used to resolve ties. This ordering was completed before applying exclusions based on age, admission type or time intervals. If a patient's first stay was ineligible, it was not replaced by a later stay. ICU-stay and hospital-admission records were linked using their corresponding identifiers. Records that could not be linked, or that had non-positive ICU or hospital intervals, were

excluded.

Patients were required to have a known age of at least 18 years. In MIMIC-IV, age was calculated by adding the difference between the hospital-admission year and anchor year to the anchor age. An anchor age of 91 was retained as an indicator of age deidentification, but calculated ages were not subsequently limited to 91. In CareVue, age was calculated in completed years between the date of birth and hospital admission. Ages shifted to 300 years or older were recorded as 91 years and marked as deidentified. These rules followed the source conventions for older patients without treating shifted dates as actual ages.

Records were required to have an admission type, and admissions classified as elective were excluded. Other named admission categories were retained according to the source mapping. No minimum ICU length of stay was required. Short stays could therefore proceed to adjudication and contribute the physiological follow-up available during the stay, without being excluded in advance on the basis of survival or stay duration.

2.2 eICU chronology and eligibility

eICU identifies ICU stays within hospital encounters and patients, but its time offsets do not allow separate hospital encounters to be ordered with certainty. The base cohort was therefore limited to patients with one hospital encounter and complete hospital-admission offsets for all ICU stays in that encounter. The ICU stay with the largest hospital-admission offset was treated as the earliest ICU admission. If more than one stay shared this position, the chronology was considered unresolved and the patient was excluded. The first ICU stay was identified before the remaining eligibility criteria were applied.

Only ICU unit types included in the predefined mapping were accepted. Step-down, virtual, mobile and test units were excluded. If an unrecognized unit type was encountered, processing was paused for review. ICU discharge was required to occur after ICU admission, hospital admission could not occur after ICU admission, and hospital discharge could not occur before ICU discharge. Patients were required to have a known age between 18 and 120 years. The source category “greater than 89 years” was recorded as 91 years and marked as deidentified.

Elective-surgery status was obtained from the APACHE admission record. Patients recorded as elective, or with conflicting elective and non-elective records, were excluded. Missing elective status was retained as unknown. Hospital discharge status was required to indicate either survival or death. Records showing ICU death followed by hospital survival were excluded as inconsistent. For patients recorded as dying in the ICU, the ICU discharge time was used as a proxy for the time of death. For other hospital deaths, the hospital discharge time was used. The source of this approximation was recorded because discharge offsets do not represent independently measured death times.

2.3 Selection summaries

Eligibility records were retained for the full screened population and stored separately from the base cohorts. Subsequent summaries reported retention at the stages of infection candidacy, sepsis adjudication, temporal follow-up and representation eligibility. Comparisons between retained and excluded groups used non-overlapping groups within each source. Age and mortality could therefore be compared at the patient level without treating repeated hourly measurements from the same patient as independent observations. The statistical procedures used for these comparisons are described in Section 10.3.

3 Suspected infection identification

3.1 Antimicrobial and microbiology evidence in MIMIC

Suspected infection was identified using an antimicrobial–culture association rule based on the Sepsis-3 operational framework [5]. Antimicrobial names were matched to predefined antibacterial and antifungal terms using word boundaries, and the recorded route had to appear in the intravenous-route mapping. Topical preparations, lock solutions, desensitization entries and other excluded formulations were removed. Records with missing start times, invalid prescription intervals or cancelled orders were not considered qualifying evidence. The available records represented prescriptions or orders and did not confirm that every dose was administered.

In MIMIC-IV, microbiology records were classified using a predefined diagnostic-test mapping, specimen identifiers and timestamps. Surveillance tests were excluded, but organism growth was not required for a diagnostic culture to qualify. Test-result rows were combined before culture events were defined. A culture was classified as cancelled only when all available result records indicated cancellation. This approach prevented separate organism or susceptibility rows from being counted as separate infection events.

CareVue did not contain equivalent test-level information for this definition, so its microbiology mapping used predefined diagnostic specimen types. Records were combined by ICU stay, specimen category and recorded time. This removed repeated result rows, although it could not determine whether records referred to the same physical specimen when specimen identifiers were unavailable. This limitation was recorded as part of the source-specific definition.

For exact antimicrobial start a and culture time c , a pair qualified when

$$a - 72 \text{ h} \leq c \leq a + 24 \text{ h}, \quad s = \min(a, c), \quad (1)$$

where s is the suspected infection time (SIT). This asymmetric window allowed a culture to occur before treatment or shortly after treatment began. All qualifying antimicrobial starts remained available for pairing. Starts were also grouped into episodes when the gap from the preceding period of coverage was no more than 24 h, but screening was not limited to the first start in an episode. Antimicrobial duration and record completeness were reviewed without requiring a minimum course of treatment.

3.2 Date-level uncertainty and candidate retention

Events recorded only by date were represented as half-open intervals from midnight on the recorded date to midnight on the following date. This rule was applied to date-only cultures in MIMIC-IV and to both antimicrobial starts and date-only cultures in CareVue. Midnight was treated as the lower bound of the possible event time rather than as the event time itself.

Antimicrobial–culture pairing produced possible SIT bounds, $[s_L, s_U]$, with an exclusive upper bound retained where applicable. A candidate was carried forward when this interval overlapped the ICU presentation window, defined as 24 h before to 24 h after ICU admission. A candidate was classified as definite only when antimicrobial–culture compatibility and membership in the presentation window held across its full timing interval. Pairs that qualified only for some possible event times were retained as possible evidence for review but could not independently determine membership in the primary cohort.

Candidates were ordered using their SIT bounds, followed by recorded event information to resolve ties. All ranked candidates were retained for SOFA adjudication. This allowed a later

candidate to be considered when the first infection pair did not have sufficient evidence of organ dysfunction. The earliest candidate meeting all adjudication criteria was then selected.

3.3 Primary eICU definition and strict-culture sensitivity

For the primary eICU definition, an intravenous antimicrobial order was paired with a diagnosis containing predefined infection terms. Diagnoses describing a ruled-out condition, medical history, prophylaxis, colonization, a non-infectious condition or a negated infection were excluded. Both records had to occur within 24 h before or after ICU admission and within 24 h of one another. For each antimicrobial start, the closest qualifying diagnosis was selected, with timing and record identifiers used to resolve ties. The antimicrobial start time was used as the SIT proxy.

This definition was used because culture–prescription evidence was not consistently available across eICU hospitals. The diagnosis time and antimicrobial start reflected documentation and treatment intent and were not interpreted as the biological onset of infection.

A separate strict-culture definition used mapped diagnostic culture sites and the same $[-72, +24]$ -h window around antimicrobial initiation. Cancelled and duplicate culture records were addressed before pairing. These candidates underwent separate SOFA association and onset adjudication. The strict-culture flag therefore required qualifying organ dysfunction associated with the candidate, rather than only the presence of a culture. This flag was used as a sensitivity annotation within the primary cohort. When this subgroup was summarized, the existing primary tensors remained aligned to the primary onset.

4 Phenotype annotation

4.1 Retrospective comorbidity burden

Charlson comorbidity components were identified from the available diagnosis codes using predefined ICD mappings [6]. Before matching, punctuation was removed from the codes and letter case was standardized. MIMIC-IV used the recorded ICD-9 or ICD-10 version, while CareVue used ICD-9. In eICU, compound diagnosis strings were separated into individual codes, and an ICD version was assigned only when the code format was clear. Ambiguous codes were not assigned to a component but were retained in the diagnosis audit.

The index was calculated without including age points. Myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, dementia, chronic pulmonary disease, rheumatic disease and peptic ulcer disease each contributed one point. Paraplegia and renal disease each contributed two points, and HIV/AIDS contributed six points. For hierarchical conditions, only the highest applicable weight was used: one or three points for mild or severe liver disease, one or two points for diabetes without or with complications, and two or six points for malignancy without or with metastases. Weights from related severity categories were not added together.

For MIMIC, components were based on diagnoses recorded across the hospital admission. For eICU, they were based on the available retrospective ICU diagnosis records. The index summarizes recorded comorbidity burden. A component value of zero means that no qualifying mapped code was found; it does not confirm that the condition was absent or that the information was available at sepsis onset. The index was therefore retained as retrospective clinical context rather than used by default as a prospective predictor.

4.2 Alternative diagnoses and restricted phenotype

Acute myocardial infarction, pulmonary embolism, pancreatitis, and trauma or burns were flagged when they appeared in either of the first two diagnosis positions in MIMIC. In eICU, the corresponding annotations were based on diagnoses marked as primary or major. Because these conditions can occur alongside infection and organ dysfunction, their presence did not automatically lead to exclusion.

A restricted-phenotype indicator identified stays that did not have these alternative diagnoses and had at least 24 h of ICU time after SIT. This indicator was retained as an annotation and recalculated for the selected infection candidate. Neither the Charlson Comorbidity Index (CCI) nor the restricted-phenotype indicator replaced SOFA adjudication. A separate chronic-organ-disease flag recorded renal disease, liver disease or congestive heart failure for the baseline-SOFA sensitivity summaries.

5 Temporal extraction and measurement harmonization

5.1 Extraction envelope and source linkage

For an ICU stay with infection candidates $j = 1, \dots, J$, the extraction window was defined as

$$[\min_j s_{Lj} - 72 \text{ h}, \max_j s_{Uj} + 48 \text{ h}]. \quad (2)$$

This window covered all candidates before one was selected. The 72-h margin before the earliest candidate included the preceding 24 h of measurements needed at the start of the 48-h SOFA association window. The 48-h margin after the latest candidate retained information needed for adjudication and construction of the post-onset tensor. In eICU, candidates from both the primary and strict-culture definitions contributed to these limits.

For MIMIC-IV, data were obtained from chart, laboratory, input, output and procedure events. CareVue used separately mapped chart, laboratory, infusion and output records. The eICU data included periodic and aperiodic vital signs, laboratory measurements, respiratory observations, nursing assessments, intake/output records and infusions. Source identifiers or labels were used to map records to clinical concepts, with reference to the relevant source code repositories [7–9]. The original value, unit, event identifier, timestamp and linkage information were retained with each mapped feature.

Laboratory measurements could be used when they occurred within the relevant hospital or ICU interval, allowing valid pre-ICU measurements to contribute to SOFA assessment. Charted ICU observations were required to occur during the ICU stay. A valid ICU record was not excluded solely because of disagreement with the recorded hospital timestamps. Treatment and procedure intervals were required to have correctly ordered endpoints and were limited to their overlap with the ICU stay, extraction window and recorded survival interval. Point measurements recorded after death were excluded.

5.2 Canonical units and numeric screening

A numeric measurement was accepted only when the source value could be interpreted, its unit was included in the unit mapping and the converted value fell within the inclusive screening limits. Measurements with unsupported units or values outside these limits were set aside rather

than clipped to the nearest boundary. The limits were used as broad data-quality checks and were not intended to represent clinical reference ranges or absolute physiological limits.

Unit conversions were defined separately for each feature. These included Fahrenheit to Celsius, kPa to mmHg, g/L to g/dL and absolute cell counts to the registered count scale. Creatinine values in $\mu\text{mol/L}$ were multiplied by 0.011312217 to obtain mg/dL, and glucose values in mmol/L were multiplied by 18.01477. Fractional differential counts were converted to percentages only when this interpretation was supported by the unit mapping. Values without recorded units were accepted only for source fields whose interpretation was specified in the configuration.

FiO₂ values from 0.2 to 1 were interpreted as fractions, while values from 20 to 100 were interpreted as percentages and divided by 100. Values that could not be assigned clearly to either scale were rejected. Vasoactive infusion rates were converted using the recorded mass or activity unit and time basis. Conversion of an absolute mass rate to $\mu\text{g/kg/min}$ required a documented weight between 20 and 400 kg. No default weight or drug concentration was assumed. The screening maxima were 10 $\mu\text{g/kg/min}$ for norepinephrine, epinephrine and phenylephrine, 100 $\mu\text{g/kg/min}$ for dopamine and dobutamine, and 1 unit/min for vasopressin.

Record validity and numeric validity were assessed separately. For example, a valid ventilator record could provide categorical evidence of respiratory support even when it did not contain a usable numeric value. Conversely, a numeric value was not included in the tensor when its timing or provenance could not be established. This distinction allowed non-numeric clinical evidence to be retained while limiting numeric arrays to accepted measurements.

5.3 Source-specific evidence restrictions

PaO₂ and PaCO₂ measurements required evidence of an arterial specimen. In MIMIC, specimen information was used when available, and records with conflicting or unknown specimen labels were excluded. In eICU, laboratory type 7 was used as a proxy for an arterial blood-gas measurement, although it did not identify the physical specimen. Groups of measurements recorded at the same time were therefore marked as inferred specimen groups. Superseded laboratory revisions were excluded, while conflicting values in the latest revision were left unresolved. Bicarbonate measured in a blood-gas panel was kept separate from the chemistry bicarbonate channel.

Infusion rates in CareVue and eICU did not have the same interval structure as those in MIMIC-IV. A recorded rate was extended to the next compatible record only when the gap was no more than 4 h. Stop markers ended the interval, and a final rate without a supported endpoint was not extended. CareVue records were linked using drug and order information, whereas eICU records required an explicit numeric rate and a unit in the drug label. Infusion-volume rates in eICU were not converted using assumed drug concentrations. When conversion depended on body weight, the weight had to be recorded in the same infusion record. Conflicting rates recorded at the same time were not used.

Glasgow Coma Scale (GCS) components were accepted only when an assessable numeric response was recorded. A verbal response marked as intubated or otherwise unassessable was not assigned a normal score. The eICU worst-day APACHE assessments were not assigned estimated timestamps. Urine output was based on mapped records of measured volume; cumulative totals, event counts and records with ambiguous times were not treated as separate volume measurements. These rules reduced the risk of counting incompatible source fields as additional observations.

5.4 Feature contract

Table M1 lists the common channel order, units, screening bounds and hourly aggregation operator. Operators for derived channels are described in Section 7.2. The urine-output bounds apply to individual accepted volume records, so the sum within an hour may exceed the upper bound for a single record. Methodology tables use the prefix M to distinguish them from the separately provided validation tables S1–S7.

6 Sepsis-3 adjudication

6.1 Assessment timeline and component scores

SOFA scores were reconstructed from the cleaned observations for the respiratory, coagulation, hepatic, cardiovascular, central nervous system and renal components [10]. Assessments were performed at the times of recorded measurements and qualifying treatments, at ICU and care endpoints, at infection-candidate boundaries and on an hourly grid across the extraction window. At an assessment time t , each component was assigned using the most severe qualifying evidence in $(t - 24\text{ h}, t)$. The lower boundary was open so that an observation exactly 24 h old was not counted in two adjacent rolling windows.

Platelet counts below 150, 100, 50 and $20 \times 10^3/\mu\text{L}$ corresponded to coagulation scores of 1, 2, 3 and 4. Bilirubin values of at least 1.2, 2, 6 and 12 mg/dL corresponded to hepatic scores of 1–4, and creatinine values of at least 1.2, 2, 3.5 and 5 mg/dL corresponded to renal scores of 1–4. GCS totals of 13–14, 10–12, 6–9 and below 6 produced central nervous system scores of 1, 2, 3 and 4. When more than one threshold applied, the highest score was used.

GCS scoring required complete and non-conflicting eye, verbal and motor responses recorded at the same time. Components from separate assessments were not combined. No retrospective adjustment was made for sedation, and an unassessable verbal response was not replaced with an assumed score. Such records therefore provided less information for GCS scoring.

6.2 Respiratory pairing and support

$\text{PaO}_2/\text{FiO}_2$ was calculated from an accepted arterial PaO_2 measurement and a compatible FiO_2 value. An FiO_2 value from the same specimen group was used when available. Otherwise, the most recent preceding charted FiO_2 within 4 h was used. If same-specimen FiO_2 records conflicted, a value from another source was not substituted, and FiO_2 values recorded after the PaO_2 measurement were not paired retrospectively.

Ratios below 400 and 300 mmHg produced scores of 1 and 2. Ratios below 200 and 100 mmHg produced scores of 3 and 4 only when invasive respiratory support was documented; without this evidence, the ratio could still contribute the corresponding lower score. Device and ventilator-mode records had to fall within the configured 4-h interval. Contradictory evidence of non-invasive support or absence of support prevented classification as invasive ventilation. In MIMIC-IV, a qualifying procedure interval could also establish invasive support. CareVue used mapped airway, ventilator-mode and device fields together with extubation records, while eICU used its corresponding respiratory-support mappings.

Table M1. Common temporal feature contract.

Feature	Unit	Inclusive bounds	Hourly operator
Heart rate	beats/min	0–300	Maximum
Mean arterial pressure	mmHg	0–300	Minimum
Respiratory rate	breaths/min	0–100	Maximum
Temperature	°C	20–45	Mean
SpO ₂	%	0–100	Minimum
GCS eye response	score	1–4	GCS assessment
GCS verbal response	score	1–5	GCS assessment
GCS motor response	score	1–6	GCS assessment
PaO ₂	mmHg	1–800	Maximum
FiO ₂	fraction	0.2–1	Maximum
PaO ₂ /FiO ₂	mmHg	1–4000	Paired minimum
PaCO ₂	mmHg	1–300	Maximum
Lactate	mmol/L	0–50	Maximum
Creatinine	mg/dL	0–50	Maximum
Blood urea nitrogen	mg/dL	0–400	Maximum
Bilirubin	mg/dL	0–100	Maximum
Platelet count	10 ⁹ /L	0–3000	Minimum
White blood cell count	10 ⁹ /L	0–1000	Maximum
Hemoglobin	g/dL	0.1–30	Minimum
pH	Dimensionless	6–8.2	Minimum
Prothrombin time	s	0–300	Maximum
Activated partial thromboplastin time	s	0–500	Maximum
Albumin	g/dL	0–15	Minimum
Potassium	mmol/L	0.5–10	Mean
Sodium	mmol/L	50–220	Mean
Glucose	mg/dL	1–5000	Maximum
Chloride	mmol/L	20–200	Mean
Urine output	mL	0–10000	Volume sum
Norepinephrine-equivalent dose	μg/kg/min	0–24.5	Concurrent peak
Invasive ventilation	Dimensionless	0–1	Support coverage
Anion gap	mmol/L	–30–100	Maximum
Bicarbonate	mmol/L	0–80	Minimum
Total calcium	mg/dL	0–20	Mean
Hematocrit	%	1–85	Minimum
Magnesium	mg/dL	0–15	Mean
Phosphate	mg/dL	0–50	Mean
International normalized ratio	Dimensionless	0.5–30	Maximum
Lymphocytes	%	0–100	Mean
Monocytes	%	0–100	Mean
Neutrophils	%	0–100	Mean
Basophils	%	0–100	Mean
Eosinophils	%	0–100	Mean
Alanine aminotransferase	U/L	0–100000	Maximum
Alkaline phosphatase	U/L	0–30000	Maximum
Aspartate aminotransferase	U/L	0–100000	Maximum
Mean corpuscular hemoglobin	pg	1–100	Mean
Mean corpuscular hemoglobin concentration	g/dL	5–60	Mean
Mean corpuscular volume	fL	30–200	Mean
Red cell distribution width (CV)	%	5–60	Mean
Red blood cell count	10 ¹² /L	0–15	Mean

6.3 Vasoactive treatment and urine coverage

Mean arterial pressure below 70 mmHg produced a cardiovascular score of 1. Dopamine at no more than 5 $\mu\text{g/kg/min}$ or any positive dobutamine rate produced score 2. Dopamine above 5 and no more than 15 $\mu\text{g/kg/min}$, or epinephrine or norepinephrine at a positive rate no greater than 0.1 $\mu\text{g/kg/min}$, produced score 3. Dopamine above 15 $\mu\text{g/kg/min}$ or epinephrine or norepinephrine above 0.1 $\mu\text{g/kg/min}$ produced score 4. Treatment had to continue for at least 60 min before it contributed to scoring. Intervals were combined separately for each drug and score threshold, and the first eligible scoring time was 60 min after continuous exposure began. Thus, treatment duration observed later was not assigned retrospectively to the start of treatment. The cardiovascular component was the higher of the pressure-based and treatment-based scores.

Urine-output records were combined at each timestamp. Identical records were counted once, while conflicting or invalid volumes left the timestamp unresolved. Separate ordinary measured outputs could be summed. Irrigant output minus irrigant input was accepted only when both values were recorded together, the result was non-negative and no ordinary urine-output record competed at the same time.

Low urine volume was used to identify severe renal dysfunction only when recording coverage was adequate. The assessment had to occur at least 24 h after ICU admission. Within the rolling window, urine records had to span at least 22 h, with the first record no more than 1 h after the start and the last record no more than 1 h before the end. No internal gap could exceed 2 h, and the window could not contain an invalid volume timestamp. When these conditions were met, a 24-h volume below 500 mL produced score 3 and a volume below 200 mL produced score 4. Otherwise, the urine-based contribution was unavailable. Renal SOFA was the higher of the creatinine-based score and the assessable urine-based score. This rule separated a documented low volume from incomplete urine recording.

6.4 Missing components, infection association and onset

Missing component scores were retained as missing, and the number of observed components was recorded for each assessment. Only when calculating the total score was an unobserved component assigned zero:

$$S(t) = \sum_{k=1}^6 \begin{cases} S_k(t), & \text{if component } k \text{ was observed,} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

For primary adjudication, baseline SOFA was assumed to be zero, so $S(t) \geq 2$ met the organ-dysfunction criterion [4]. This assumption was stored separately from the observed component evidence and was not treated as evidence that chronic organ dysfunction was absent.

For an exact SIT s , a qualifying assessment during the ICU stay had to occur in $[s - 48 \text{ h}, s + 24 \text{ h}]$. When SIT was represented by an interval, definite compatibility required

$$t \in [s_U - 48 \text{ h}, s_L + 24 \text{ h}]. \quad (4)$$

This interval is the intersection of the association windows allowed by the candidate's timing bounds. Assessments that fell only within the wider possible window were retained for review but did not establish primary sepsis. A qualifying candidate also required definite infection pairing, definite presentation timing, usable SOFA evidence and an onset within the recorded follow-up period.

Let t^* be the first qualifying SOFA assessment for a candidate. Its onset bounds were

$$o_L = \max(s_L, t^*), \quad o_U = \max(s_U, t^*). \quad (5)$$

The first qualifying candidate in the ranked list was selected. Its lower onset bound was used to align the hourly tensor, while the upper bound and endpoint convention were retained to describe timing uncertainty. The upper onset bound was limited by ICU discharge or recorded death. Availability of complete 24-h follow-up was assessed from the upper bound, providing a conservative classification when onset time was uncertain. Candidate- and stay-level audit records distinguished unavailable evidence, uncertain timing, SOFA below the threshold and onset outside the recorded follow-up period.

6.5 Measured-baseline sensitivity

A sensitivity annotation compared the highest measured SOFA before SIT within the association period with the highest score from SIT through the following 24 h. An increase of at least two points met the measured-baseline sensitivity criterion. Each assessment required at least one observed component. If either the pre-SIT or post-SIT evidence was unavailable, the comparison was classified as unknown. Across candidates, a stay was positive when at least one candidate met the criterion, unknown when none met it but at least one comparison was unavailable, and negative otherwise.

This analysis examined the effect of the zero-baseline assumption used in the primary definition. The pre-SIT maximum was not considered a pre-morbid baseline, and the CCI was not converted into a baseline SOFA score. The stored baseline PaO₂/FiO₂ value was similarly defined as the lowest valid paired ratio during the 24 h strictly before the released onset and was retained as retrospective context. These sensitivity annotations did not change the primary tensor population.

7 Observed tensor construction

7.1 Hourly exposure and structural unavailability

For an onset anchor o and follow-up endpoint f , the observation period ended at $e = \min(o + 24 \text{ h}, f)$. Each hour $h = 0, \dots, 23$ represented $[o + h \text{ h}, o + (h + 1) \text{ h})$, limited by e . Exposure was calculated in seconds as

$$E_{ih} = \max\{0, \min(o_i + (h + 1) \text{ h}, e_i) - (o_i + h \text{ h})\}. \quad (6)$$

The time difference in this expression was evaluated in seconds. When follow-up ended partway through the final hour, that hour was retained with its actual exposure. An event recorded exactly on an hourly boundary was assigned to the following hour, while events at the final observation endpoint were excluded.

When hospital death was recorded without a corresponding death time, the patient and mortality outcome were retained, but temporal exposure was not assigned. This prevented measurements from being placed within an assumed survival interval that might extend beyond death. For eICU, the recorded death-time proxy described above was used.

The observed tensor had dimensions $N \times 24 \times 50$ and was stored using 64-bit floating-point values. Missing cells were represented as NaN. A positive observation count identified a cell

containing an observed finite value, whereas zero exposure indicated that the hour was structurally unavailable. A missing measurement during an available hour could therefore be distinguished from an hour occurring after follow-up had ended. The same distinction was maintained during subsequent imputation.

7.2 Direct and derived hourly values

Direct measurements were aggregated using the minimum, maximum or mean specified in Table M1. Exact duplicates with the same source, item, specimen, time and value were removed before aggregation. The operator for each channel was defined in advance and was not chosen according to outcomes or agreement between databases. Minimum and maximum operators retained selected hourly extremes, while the mean summarized channels intended to represent an average value. These summaries do not preserve every original measurement.

The hourly GCS channels were taken from the complete assessment with the lowest total score, with the latest timestamp used to resolve ties. If the hour contained no complete assessment, the latest partial assessment was used and unavailable components remained missing. The PaO₂/FiO₂ channel used the lowest valid paired ratio recorded within the hour and was not recalculated from the separately aggregated PaO₂ and FiO₂ channels. Urine output was calculated by summing accepted volumes. If any urine timestamp within the hour was unresolved, the hourly volume was withheld. An hourly urine sum did not by itself establish sufficient recording coverage for the separate 24-h SOFA criterion.

Norepinephrine-equivalent dose (NEQ) used the five extracted agents in the configured conversion [11]:

$$\text{NEQ}(t) = \text{NE}(t) + \text{Epi}(t) + 0.1 \text{ Phenyl}(t) + 0.01 \text{ Dopa}(t) + 2.5 \text{ Vaso}(t). \quad (7)$$

Vasopressin was expressed in units/min, while the other agents were expressed in $\mu\text{g/kg/min}$. Infusion interval endpoints divided each hour into segments. Accepted rates active at the same time were summed within each segment, and the highest segment value was retained for the hour. Dobutamine was not included in NEQ, and vasoactive agents that were not extracted were assigned no contribution. The 60-min treatment-duration requirement used for SOFA scoring was not applied to this descriptive dose channel.

Identical overlapping rates for the same drug were counted once rather than treated as separate additive infusions. An hour was considered unresolved when simultaneous rates conflicted or when an active rate could not be interpreted. If an infusion endpoint was unresolved, the NEQ channel was withheld for the stay because its coverage could not be determined. The absence of treatment records was not interpreted as a zero dose.

Invasive ventilation was assigned a value of one when an interval of documented, non-conflicting invasive support overlapped the hour for a positive duration. A value of zero required explicit evidence of no support or non-invasive support throughout the available portion of the hour. Otherwise, the value remained unknown. Separate coverage fields for interval-based features recorded how much of each hour was supported by available evidence. Hourly observation coverage and cell provenance for the three databases are shown in Supplementary Fig. S1.

7.3 Static context and descriptive summaries

Static arrays included age at admission and the mapped demographic and admission categories. Integer category codes represented entries in a fixed vocabulary and did not imply an ordered

clinical scale. Unknown or unmapped categories remained missing, and the original labels were retained in the metadata. CCI, assumed baseline SOFA and retrospective pre-onset $\text{PaO}_2/\text{FiO}_2$ were stored as separate contextual variables so that they were not treated as admission-time predictors.

Clinical distribution summaries were based on one mean for each patient and feature, calculated from that patient's observed hourly values. Medians and other quantiles were then calculated across these patient-level means, so each patient with observed data contributed once. Figure 2 displayed eight predefined physiological features representing vital signs, perfusion, renal, hepatic, coagulation and respiratory measurements. The same summary method was applied to all 50 channels in the supplementary table. The displayed features were selected for presentation and were not chosen using outcome information.

Relationships between features were summarized using pairwise Spearman correlations between observed patient-level means. Average ranks were used for ties, and each correlation required at least three patients with both features observed. Correlations were left unavailable when either feature was constant or when too few paired observations were present. Study-support surfaces counted patients with at least k features observed in at least h hourly bins, using the full clinical cohort as the denominator. Availability based on observed values was reported separately from availability after permitted filling. These summaries were descriptive and did not alter feature eligibility or cohort membership.

8 Gap-aware SAITS imputation

8.1 Patient partitions and training-only preprocessing

Separate SAITS models were trained for the three databases [12]. Within each source, patients were ordered using the SHA-256 digest of a string combining seed 42 with the subject identifier. The first $\lfloor 0.70N \rfloor$ patients were assigned to training, the next $\lfloor 0.15N \rfloor$ to validation and the remaining patients to testing. Assignment did not depend on outcomes or the original row order. The hash-ordered validation partition was divided into two halves: the first was used for early stopping and the second for method selection. The same partitions were used for all subsequent representations.

The three GCS components, $\text{PaO}_2/\text{FiO}_2$, urine output, NEQ and invasive ventilation were treated as observed-only channels. Missing values in these channels could indicate that an assessment was not performed, pairing was unresolved or treatment coverage was unavailable. They were therefore not imputed as ordinary continuous variables. Each remaining channel required at least 50 finite observations from at least 25 patients in the training partition. Patients with no observed values in any eligible continuous channel provided no model context and did not receive imputed values.

The transformation, mean, standard deviation and physical-unit median for each feature were estimated from the training observations. ALP, ALT, AST, basophil percentage, bilirubin, BUN, creatinine, eosinophil percentage, lactate, monocyte percentage and WBC were transformed using $\log(1 + x)$ before standardization. Other modeled channels were standardized without a preceding transformation. When a feature had zero standard deviation, the scale was set to one and the feature was marked as constant. Validation and test observations were not used to estimate these quantities.

8.2 Architecture and gap-aware training

The model used a two-block SAITS architecture with two attention layers in each block, a model dimension of 256, a feed-forward dimension of 128, four attention heads and key and value dimensions of 64. Diagonal attention masking was enabled. Fully connected dropout was set to 0.1 and attention dropout to zero. The observed-reconstruction and masked-imputation losses were assigned weights of one and two, respectively. Models were trained with the specified PyPOTS implementation using batches of 64, up to 60 epochs and an early-stopping patience of eight epochs. The default Adam optimizer used a learning rate of 10^{-3} , moment coefficients (0.9, 0.999), a numerical stabilizer of 10^{-8} and zero weight decay, without a learning-rate scheduler [16]. SAITS used information in both temporal directions within the retrospective 24-h sequence and was not interpreted as a causal forecasting model.

For each training example, artificial missingness was sampled independently from three patterns with equal probability:

- Point masking removed 20% of eligible observed cells, rounded down. At least one target and one observed context cell were retained whenever possible.
- Block masking removed observed cells from all modeled features within one continuous 6-h interval. The selected interval had to contain at least one target and leave observed context outside the interval.
- Whole-channel masking removed all observed values from one channel while retaining observations in at least one other channel. Channel-sampling weights were proportional to the inverse square root of training-patient coverage.

If a structured masking pattern produced no usable target, point masking was used instead. This mixture represented scattered missing values, interruptions in recording and complete absence of a channel, rather than training only on isolated withheld measurements.

For a target mask H and standardized prediction error r_{ihf} , the feature-balanced Huber criterion was

$$\mathcal{L}(H) = \frac{1}{|\mathcal{F}_H|} \sum_{f \in \mathcal{F}_H} \frac{\sum_{i,h} H_{ihf} \ell_1(r_{ihf})}{\sum_{i,h} H_{ihf}}, \quad \ell_1(r) = \begin{cases} r^2/2, & |r| \leq 1, \\ |r| - 1/2, & |r| > 1. \end{cases} \quad (8)$$

Here, $\mathcal{F}_H$ is the set of features with at least one target in the batch. Errors were averaged within each feature before being averaged across features, reducing the influence of densely recorded variables. The Huber transition reduced the effect of large standardized errors. Early stopping used the corresponding feature-balanced absolute error calculated on fixed validation masks. Structurally unavailable hours had zero observed and target masks and were neither scored nor filled. The SAITS implementation did not include a separate attention-padding mask for structural hours.

8.3 Fixed holdouts and feature-specific method selection

The early-stopping holdouts included no more than 512 patients for each point- and block-masking scenario and no more than 16 patients per feature for whole-channel masking. The method-selection and test whole-channel holdouts included no more than 128 patients per feature. Each whole-channel example hid one channel, so the same patient could contribute separate examples for different features. Point and block masks were generated deterministically from the

dataset role, masking scenario and subject identifier. Patient indices and masks were saved so that evaluation used fixed targets.

SAITS was compared with two alternatives: the training median and forward filling. Forward filling used only earlier available observations from the same patient and feature. The training median was used before the first observation and when the channel was entirely missing. Predictions were inverse-transformed before comparison so that all errors were calculated in physical units.

For feature f , scenario s and method m , let E_{fsm} be cell-weighted mean absolute error (MAE) on the fixed validation targets. Equal-scenario normalized error was

$$Q_{fm} = \frac{1}{3} \sum_s \frac{E_{fsm}}{\max(E_{fs,\text{median}}, 10^{-12})}. \quad (9)$$

The lower-scoring rule between the median and forward filling was selected as the baseline, with ties resolved in favor of the median. Before evaluating SAITS, predictions outside the channel's screening bounds were replaced by the selected baseline. SAITS was accepted only when each scenario contained at least 25 targets from ten patients, no more than 5% of predictions were outside the screening bounds in any scenario, and

$$E_{f,s,\text{SAITS+fallback}} \leq 1.05 \min(E_{f,s,\text{median}}, E_{f,s,\text{forward}}), \quad \text{for every } s, \quad (10)$$

$$Q_{f,\text{SAITS+fallback}} \leq 0.99 Q_{f,\text{baseline}}. \quad (11)$$

The baseline score was also required to be positive. A tolerance of 10^{-12} was used for numerical comparisons where specified. When validation evidence was insufficient, the training median was selected. These criteria required an overall improvement across the three missingness scenarios while limiting deterioration in any individual scenario.

After the model checkpoint and feature-specific methods were fixed, performance was evaluated on the three test scenarios. In the final application, only naturally missing eligible cells within periods of positive exposure were filled, and only for patients with observed continuous-channel context. Recorded values were left unchanged. SAITS predictions outside the screening bounds were replaced by the previously selected baseline rather than clipped. Cell-level codes distinguished unfilled, observed, SAITS-imputed, forward-filled and median-filled values. A separate mask identified cells where the baseline was used because a SAITS prediction exceeded the bounds.

8.4 Patient-level validation and temporal distortion

Statistical validation reapplied the saved test masks and compared their target counts and cell-weighted errors with the stored imputation report. For inference, errors were first averaged within each patient. If H_{if} represents a patient's hidden targets for a given feature and scenario, the primary estimate was

$$\text{MAE}_f = \frac{1}{n_f} \sum_{i=1}^{n_f} \frac{1}{|H_{if}|} \sum_{h \in H_{if}} |\hat{x}_{ihf} - x_{ihf}|. \quad (12)$$

This estimate gives each patient equal weight and therefore differs from the cell-weighted error used during method selection. Signed errors and paired differences in patient-level MAE relative to each baseline were also calculated. A negative difference indicated lower error for the final

method. Supplementary Fig. S2 shows these paired differences and their uncertainty intervals for selected features.

Tail analyses used the 10th and 90th percentiles of the feature’s training observations. Additional strata described the amount of remaining observed context and the duration of contiguous gaps: fewer than 24, 24–95 or at least 96 context cells, and gaps of 1–2, 3–6 or 7–24 h. Temporal-change error compared reconstructed and observed differences between adjacent hours when both original endpoints were measured and at least one had been hidden for evaluation. Variability error was the absolute difference between reconstructed and original within-patient standard deviations and required at least three originally measured hours and one hidden value. These analyses used withheld measurements with known values and did not assign assumed ground truth to naturally missing cells.

9 Cross-database temporal representation and domain transport

9.1 Shared evidence and reference scaling

A feature was included in the shared encoder only when it was observed in at least 40 training patients and at least 5% of training patients in each database. At least 12 features had to meet these criteria. A patient was eligible for representation construction when at least 12 cells were observed across at least three shared features. These requirements applied only to the shared representation; patients who did not meet them and all 50 clinical channels remained in the source-specific datasets.

MIMIC-IV was used to define the reference scaling and train the encoder. For each feature, the median m_f and interquartile range q_f of the training observations defined

$$\tilde{x}_{ihf} = \text{asinh}\left(\frac{g_f(x_{ihf}) - m_f}{q_f}\right). \quad (13)$$

Here, $g_f(x) = \log(1 + x)$ for lactate, creatinine, BUN, bilirubin, WBC, ALT, AST, ALP, PT, aPTT and INR, while $g_f(x) = x$ for the other features. If the interquartile range was below 10^{-6} , the larger of the training standard deviation and one was used as the scale. Age was transformed using the same median and interquartile-range approach followed by the inverse hyperbolic sine. This transformation reduced the numerical influence of extreme values while preserving their order and using the same reference scale for all databases.

9.2 Temporal convolutional encoder

The temporal convolutional network (TCN) used the ordered hourly sequence to retain information about within-day timing [13]. At each hour, the encoder received $4F + 2$ inputs for F shared features: scaled values multiplied by their method weights, indicators of observed values, method weights, transformed observation counts, fractional hourly exposure and the relative hour $h/23$. Observation counts were transformed as $\log(1 + c)/\log(11)$. The weights assigned to unfilled, observed, SAITS-imputed, forward-filled and median-filled values were 0, 1, 0.5, 0.35 and 0.2, respectively. These fixed design weights represented value provenance and were not calibrated probabilities of uncertainty. Inputs outside the follow-up period were set to zero.

The network contained three one-dimensional convolutional layers with 16 hidden channels, a kernel width of three and dilations of one, two and four. Symmetric padding preserved the

sequence length. Each layer applied a GELU activation followed by the exposure mask, and the convolutions were non-causal. Hidden states were averaged within four consecutive 6-h blocks using exposure as the weight. The four block summaries were then concatenated in temporal order to produce a 64-coordinate representation. During training, a width-one convolution reconstructed the feature channels.

Encoder training required at least 100 eligible MIMIC-IV training patients and 30 eligible validation patients. During training, 25% of observed shared cells were randomly withheld and reconstructed from an observed-only input view. Generated values were removed from this view so that previous imputation could not reveal a withheld target. Reconstruction used a feature-balanced smooth-L1 loss with a transition value of one. A second loss term encouraged agreement between representations derived from the completed and observed-only inputs:

$$\mathcal{L}_{\text{TCN}} = \mathcal{L}_{\text{reconstruction}} + 0.05 \text{ MSE}(z^{\text{completed}}, \text{stopgrad}(z^{\text{observed}})). \quad (14)$$

The observed-only representation in this term used the available original measurements and was separate from the masked input used for reconstruction. AdamW used a learning rate of 10^{-3} , weight decay of 10^{-4} , batches of 128 and gradient-norm clipping at five. Training continued for no more than 40 epochs, with a patience of six epochs and a minimum validation improvement of 10^{-6} . The model checkpoint was selected by reconstruction loss on fixed validation masks withholding 25% of observed cells.

Each latent coordinate was then centered and scaled using eligible MIMIC-IV training representations. Standard deviations close to zero were replaced by one. The fitted transformations and encoder were frozen and applied to every database. Observed-only representations were also saved, together with fixed perturbations that removed either 25% of observed cells or one continuous 6-h block. Clinical anchors were recalculated after each perturbation. A block-summary comparator retained the ordered 6-h feature means and observation fractions without training an additional temporal encoder.

9.3 Clinical anchors and preservation probes

Transport anchors included age and the observed means during hours 0–5 for heart rate, mean arterial pressure, creatinine, lactate, platelets and FiO₂. These means were calculated after applying the reference transformation. A prototype pair required at least two clinical anchors measured in both patients. Age contributed to the discrepancy calculation when available but did not count toward this minimum.

Preservation probes were defined for eight features: heart rate, mean arterial pressure, respiratory rate, creatinine, lactate, platelets, bilirubin and FiO₂. Each feature contributed an early mean and the change from that early mean to the mean during hours 18–23, producing up to 16 targets. Ridge regressions with a penalty of ten were fitted from eligible MIMIC-IV training representations to the observed targets. Each probe required measurements from at least 40 training patients and sufficient variation in the target. The probe coefficients and the 10th- and 90th-percentile tail cutoffs from the training targets were fixed before validation in the target databases. These regressions assessed whether measured clinical information was retained; mortality was not used as a fitting target.

9.4 Unbalanced transport between training prototypes

For each target database, up to 1,024 eligible training patients were sampled without replacement as target prototypes, separately from the reference prototypes. Each prototype was an actual

patient representation. For target prototype i and reference prototype j , the latent mean-squared distance was $d_{ij} = \|z_i - z_j\|_2^2/64$. The anchor discrepancy a_{ij} was the mean-squared difference across anchors available for both patients. A pair was considered admissible only when at least two clinical anchors were shared and $a_{ij} \leq 9$, equivalent to an anchor root-mean-square discrepancy no greater than three.

For admissible pairs, the cost was

$$C_{ij} = \frac{d_{ij}}{\max(\text{median}_{\text{admissible}} d, 10^{-6})} + a_{ij}. \quad (15)$$

Pairs that were not admissible were assigned a cost of 1,000. Rows and columns without any admissible pair were removed before optimization. Their original uniform prototype masses were retained, and the masses of the remaining prototypes were not renormalized. The unbalanced formulation allowed some mass to remain unmatched when corresponding support was absent from the two training distributions [14, 15].

With target and reference prototype masses a and b , the non-negative plan P minimized

$$\langle P, C \rangle + \varepsilon \text{KL}(P \| ab^\top) + \rho \text{KL}(P \mathbf{1} \| a) + \rho \text{KL}(P^\top \mathbf{1} \| b), \quad (16)$$

where the generalized KL divergence includes the mass terms, $\text{KL}(u \| v) = \sum_k [u_k \log(u_k/v_k) - u_k + v_k]$. The stabilized unbalanced Sinkhorn solver regularized the plan against the product measure using KL divergence [17]. The tested values were $\varepsilon \in \{0.05, 0.2\}$ with $\rho = 1$, a maximum of 3,000 iterations and a stopping tolerance of 10^{-7} . An accepted plan required a finite final error no greater than 10^{-6} , finite non-negative entries and no numerical warning.

A target prototype was accepted only when its transported row mass was at least 25% of its original mass and no more than 10^{-6} of that mass was assigned to incompatible pairs. At least 30 target prototypes had to be accepted. The barycentric residual for an accepted prototype was

$$\Delta_i = \frac{\sum_j P_{ij} z_j}{\sum_j P_{ij}} - z_i. \quad (17)$$

Reliability was defined as the ratio of transported mass to original mass, capped at one. The residuals represented local corrections supported by the accepted correspondences between training prototypes.

9.5 Supported local correction and covariance comparator

Reference and target support radii were estimated separately as the 99th percentile of the second-nearest prototype mean-squared distances among training patients, with a minimum value of 10^{-6} . The second-nearest distance was used to avoid zero self-distances for patients selected as prototypes. A target patient was considered supported only when a clinically compatible reference prototype fell within the reference radius and an accepted, compatible target prototype fell within the target radius.

For a supported patient, up to eight of the nearest compatible target prototypes within the radius contributed to the correction. Their weights were proportional to reliability divided by distance plus 10^{-6} and were then normalized to sum to one. The weighted residual was attenuated by $\exp(-d_{\min}/r_T)$, where $d_{\min}$ was the distance to the nearest eligible target prototype and r_T was the target radius. Candidate correction strengths were 0.125, 0.1875, 0.25 and 0.5. After the strength was applied, the correction was rescaled when necessary so that its coordinate-wise

root-mean-square magnitude did not exceed 0.35. Patients without adequate support retained their original representation unchanged.

A covariance-based comparator used Ledoit–Wolf covariance estimates fitted to the target and reference training representations. Target vectors were centered, whitened using the target covariance and recolored using the reference covariance before the reference mean was restored. Eigenvalues were bounded below by 10^{-6} . This correction used a strength of 0.25 and the same magnitude limit. Its support rule required a compatible reference prototype and proximity to a target prototype, but target-anchor compatibility was not required when calculating the target distance.

9.6 Validation gates and locked evaluation

Candidate mappings were selected separately for each target database using its existing validation partition. Selection required at least 30 eligible validation patients, with supported corrections applied to at least 20% of them. The alignment assessment additionally required at least 30 supported patients with scorable distances and a reduction in their mean distance to the nearest clinically compatible reference training prototype. Stability had to improve by at least 2% under both point removal and 6-h block removal, based on observed-only representations and their perturbed versions across eligible validation patients.

Clinical preservation required at least four valid probe dimensions, each with measurements from at least 20 target patients. For every assessable probe, the adapted MAE could not exceed 1.05 times the original MAE plus 0.02 in transformed probe units. The same limit was applied to measured tails containing at least ten patients. For change probes, the adapted and unadapted predictions had to agree in sign for at least 95% of informative patients when at least 20 such patients were available and the original prediction magnitude exceeded 0.05. This rule assessed preservation of the predicted direction and did not evaluate whether that direction was biologically correct.

Pairwise geometry was evaluated using 2,000 sampled patient pairs. Self-pairs and pairs with near-zero original distance were excluded. The 10th percentile of the adapted-to-original distance ratios had to be at least 0.85, and the 90th percentile could not exceed 1.15. Candidates meeting all criteria were ranked by their relative improvement in supported reference alignment. Ties were resolved by preferring the lower correction strength and then the candidate name. If no candidate passed every criterion, the identity mapping was selected. The selected mapping, the best accepted transport comparator and the covariance comparator were retained for evaluation.

The selected mapping was applied to the test partition without refitting or reselection. Patient-paired differences were used to summarize reference distance, instability under recording perturbations and probe error. Probe-tolerance residuals were calculated as adapted absolute error minus 1.05 times original absolute error minus 0.02. Pointwise intervals indicated whether the specified engineering limit was met, exceeded or unresolved; they were not clinical-equivalence tests. Geometry ratios were treated as descriptive because the sampled patient pairs were not independent. Evaluation records retained information about earlier test exposure when dataset preparation or validation procedures were revised. Supplementary Fig. S3 summarizes these transport diagnostics.

9.7 Reference atlas and observed trajectories

Two-dimensional principal components and eight K-means reference groups were fitted using only eligible MIMIC-IV training representations. PCA used the full solver, while K-means used ten initializations and the fixed atlas seed. The fitted objects were then used to project patients and assign groups in every database. Original and adapted representations used the same coordinate system. Patients who were not eligible retained missing coordinates and a missing-group indicator rather than being removed.

Hourly summaries for each group were calculated from the original observed measurements. Medians and interquartile ranges were reported together with counts of available and observed patients. Generated values were not used to construct the clinical trajectories describing the groups. The groups provided an organization of the retrospective representations for visualization and dataset analysis and were not defined as clinical sepsis phenotypes.

10 Software and computational environment

10.1 Execution environment and saved artifacts

The pipeline was run with Python 3.10.13 on a workstation with a four-core CPU, 8 GB of RAM and an NVIDIA GTX 1650 Ti GPU with 4 GB of video memory. The Docker configuration used the PyTorch 2.2.1 runtime image with CUDA 12.1. Direct dependencies were fixed at NumPy 1.26.4, SciPy 1.13.1, pandas 2.2.2, PyArrow 16.1.0, DuckDB 1.1.3, scikit-learn 1.5.2, joblib 1.4.2, threadpoolctl 3.6.0, PyTorch 2.2.1, PyPOTS 1.5, POT 0.9.4, Matplotlib 3.9.2 and Pillow 10.2.0. The environment specification also included the notebook packages available in the working environment.

The three source pipelines were organized into parallel numbered stages under `src/01_mimiciv_datasets`, `src/02_mimic3_datasets` and `src/03_eicu_datasets`. Shared clinical definitions were stored in `src/common`. Representation learning and transport were implemented in `src/04_atlas_datasets`, statistical validation in `src/05_statistical_analysis` and figure construction in `src/06_visualization`. Source mappings, thresholds and model settings were stored with the versioned code rather than reconstructed from the resulting figures.

Source imputation used seed 42, atlas construction used seed 20260916 and statistical resampling used seed 20260917, with deterministic keys assigned to individual comparisons. Training set the Python, NumPy and PyTorch random seeds and disabled cuDNN benchmarking. Deterministic operations were requested when supported. Model weights, preprocessing statistics, feature order, patient partitions, holdout masks and selected methods were saved so that evaluations could reuse the fitted artifacts. These measures documented the computational procedure but did not assume that different hardware or library versions would produce bitwise-identical models.

10.2 Integrity and provenance checks

Source-level quality control checked cohort identity, row alignment, tensor dimensions, feature order, unit definitions, exposure boundaries, observation counts, mortality linkage and cell-level provenance. It also confirmed that imputation did not alter observed values, structurally unavailable hours remained unfilled and generated values were limited to eligible channels

and patients. Checks for derived features compared interval coverage with the underlying source evidence. Atlas checks confirmed that patient partitions were inherited correctly, fitting used training data only, eligibility rules were applied, mappings could be replayed, corrections remained within their bounds and unsupported representations were unchanged.

The statistical and visualization stages used fixed artifacts and linked their inputs and code through hashes. When a dependency changed, the corresponding receipt was invalidated and the affected stage and dependent stages had to be regenerated. Figure quality control also compared the expected aggregate and panel files with the visualization manifest. These checks assessed consistency and provenance and did not depend on whether the clinical or statistical results were favorable.

The release exporter checked alignment between the clinical and representation products, retained source-qualified patient keys and expressed event metadata as time offsets. The release manifest, data dictionary, cohort-flow table, Croissant metadata and SHA-256 checksums described the contents and supported file-integrity checks. The source code and environment definitions were provided at <https://github.com/MrRajat1809/ClinicalTensorSepsis>.

10.3 Statistical inference

Statistical analyses were performed after the dataset and methods had been fixed. They did not remove patients, change imputation rules or refit the transport mappings. Inferential summaries used one independent contribution per patient. Repeated hours, multiple holdout targets and repeated examples from the same patient were first reduced to the patient-level quantity being estimated. Comparisons between retained and excluded patients or between data partitions used non-overlapping groups within each source. Differences between sources were treated as descriptive because shared variable names did not make the populations exchangeable or establish independence between the MIMIC versions.

Continuous means and paired mean differences were estimated using 2,000 bootstrap replicates sampled at the patient level, with percentile 95% confidence intervals. For differences between independent groups, each group was resampled separately. Standardized mean differences were calculated by dividing the mean difference by the square root of the average of the two sample variances. An interval required at least 20 contributing patients, or at least 20 patients in each independent group, and at least 90% finite bootstrap estimates. When these conditions were not met, the interval was reported as unavailable rather than replaced with a zero-width interval. The intervals were pointwise and did not include an adjustment for simultaneous coverage.

Binary proportions used Wilson intervals, including for nonempty groups with no observed events. Risk differences used Newcombe intervals derived from the two Wilson intervals. Independent continuous groups were compared using Welch's t test and the two-sided Mann–Whitney test, while independent binary groups were compared using Fisher's exact test. Paired continuous differences were assessed with a one-sample t test applied to the patient-level differences and a two-sided sign test applied to non-zero differences. These tests addressed different null hypotheses: mean differences for the t tests, distributional differences for the rank tests and direction among non-tied pairs for the sign test.

Benjamini–Hochberg adjustment was applied within named test families defined by database, question, population, scenario, comparator and test type. Every tested feature and stratum in a family was included in the adjustment. Effect sizes, denominators and intervals were reported with the test statistics and adjusted values. Statistical significance was not used as a quality-control criterion or as a reason to change the released dataset.

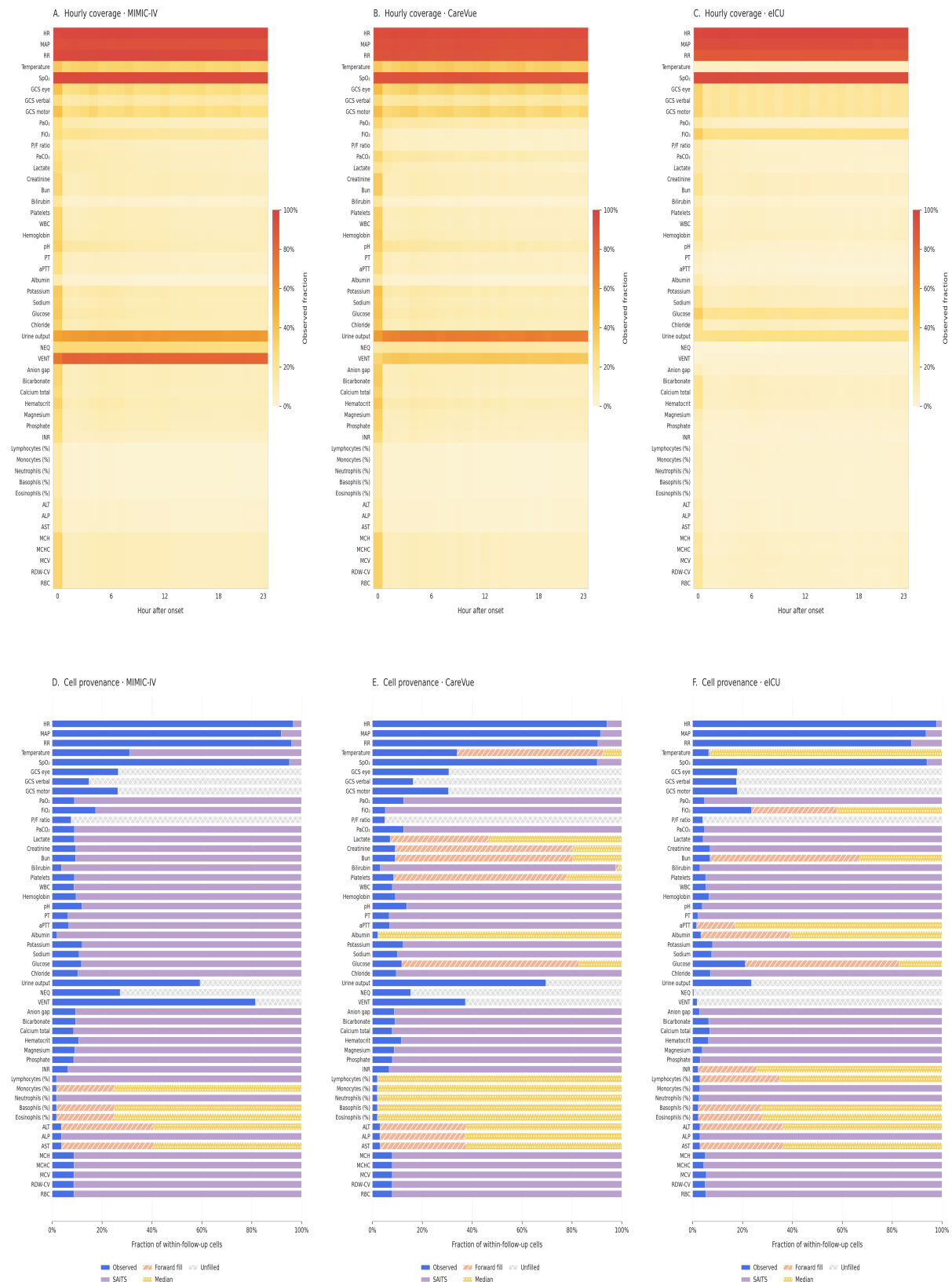
10.4 Measurement availability, site heterogeneity and sensitivity analyses

Measurement audits reported raw event-row rejection counts separately from patient-level coverage. Because a patient could have records associated with more than one rejection reason, patient counts across reasons were not added as if they were mutually exclusive. Feature availability included the proportion of clinical patients with at least one observed value, the fraction of positive-exposure hours observed for each patient and the hourly observation proportion among patients still under follow-up. Patients without temporal exposure remained in the cohort denominators and were identified separately when an exposure-based proportion could not be calculated. Fractions of observed and generated cells were reported separately.

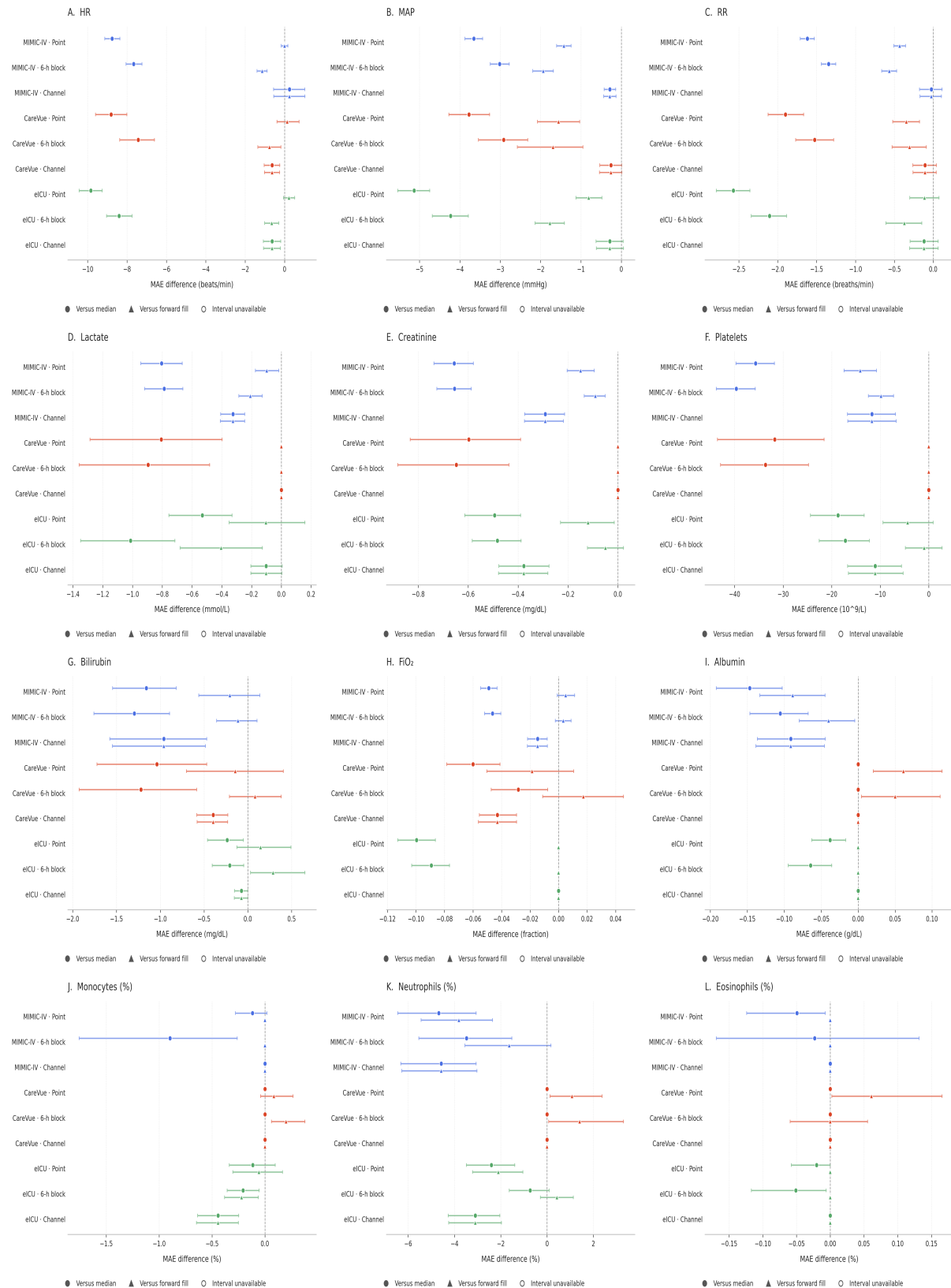
eICU hospital summaries used the recorded hospital identifiers. A hospital required at least 30 cohort patients to be included in the specified inferential site comparisons. Tests of continuous features additionally required at least 20 measured patient-level summaries in each hospital. Welch's one-way analysis of variance compared hospital means, the Kruskal–Wallis test compared distributions and the median-centered Brown–Forsythe procedure compared variances. Effect summaries included the between-hospital variance fraction, rank-based epsilon-squared and the range of hospital standard deviations. Tests involving small or constant groups were marked as unavailable.

Hospital-level observation availability was represented by one available or unavailable indicator for each patient and feature. Association with hospital was summarized using a chi-square test and Cramér's V . The asymptotic p value was not reported when any expected cell count was below five. These site effects were unadjusted and could reflect both differences in case mix and differences in recording practice. The eICU data partitions were defined at the patient level and did not hold out entire hospitals.

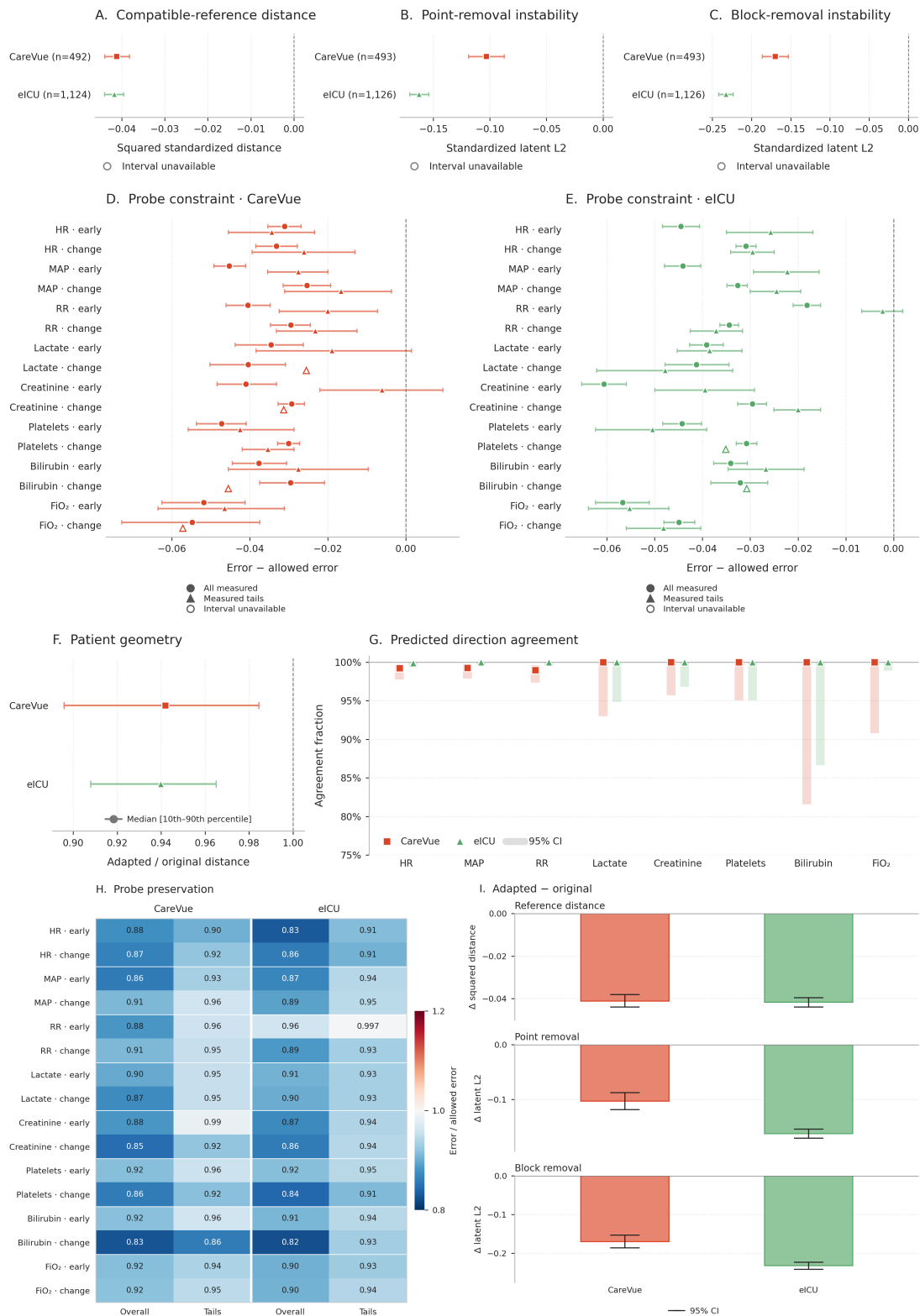
Readiness summaries identified unsupported or constant channels, observed patient coverage below 5% and features for which generated values made up more than 80% of finite completed cells. They also identified features that were absent from individual eICU hospitals meeting the 30-patient threshold. These indicators described properties relevant to future study design and did not remove features or patients. Sensitivity summaries compared the zero-baseline and measured-baseline SOFA annotations, described strata with recorded chronic organ disease and retained the eICU strict-culture subgroup with its original primary-tensor alignment. Sensitivity analyses were not used to choose a different primary cohort after reviewing the results. Supplementary Fig. S4 summarizes hospital availability, differences between patients with and without transport support, partition and cohort comparisons, and baseline-SOFA sensitivity.



Supplementary Figure S1. Hourly observation coverage and cell provenance. A–C, observed fractions for all 50 features in MIMIC-IV, CareVue and eICU, respectively, among patients with positive exposure in each hour. D–F, corresponding fractions of within-follow-up cells that were observed, filled by SAITS, forward filled, median filled or left unfilled. Structurally unavailable hours are excluded from these denominators.



Supplementary Figure S2. Patient-level imputation error differences. A–L, paired mean absolute error (MAE) differences for 12 features under point, 6-h block and whole-channel masking in the held-out test partitions. Differences are the final feature-specific rule minus the training-median baseline (circles) or forward filling (triangles), in the displayed physical units. Negative values indicate lower error for the final rule. Bars show 95% patient-bootstrap confidence intervals; open markers indicate unavailable intervals.



Supplementary Figure S3. Transport and preservation diagnostics in target test patients. A–C, adapted-minus-original differences in compatible-reference distance and instability after point or block removal. D–E, clinical-probe error minus its allowed error, defined as 1.05 times original error plus 0.02, for all measured values and measured tails. F, median and 10th–90th percentiles of adapted/original patient-distance ratios. G, agreement with original predicted change directions. H, probe error divided by allowed error. I, summary of the differences in A–C. Intervals are 95% patient-bootstrap intervals, except Wilson intervals in G and percentile ranges in F. Open markers indicate unavailable intervals.



Supplementary Figure S4. Hospital variation, cohort composition and sensitivity analyses. A, feature availability in eICU hospitals with at least 30 cohort patients. B, association between hospital and availability, expressed as Cramér's V ; open markers denote unavailable tests. C–D, mortality-risk and mean-age differences between transport-supported and unsupported eligible patients, with 95% Newcombe and patient-bootstrap intervals, respectively. E, between-hospital variance fractions. F, validation- and test-versus-training rank-biserial differences. G, standardized cohort differences relative to MIMIC-IV. H, primary-cohort fractions meeting, not meeting or lacking evidence for the measured-baseline SOFA sensitivity criterion.

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